

Supplementary material: risk assessment worksheets

Both tables use the rating scheme of Table I of the paper: impact ratings S (safety), O (operational), F (financial) and P (privacy) on the scale Severe 10, Major 7, Moderate 4, Negligible 1; impact value $I = (5S + O + 0.5F + 0.25P)/4$; feasibility from the attack vector (L local 4, A adjacent 7, N network 10); risk score $R = 1 + I \times F$. Treatment: Red. risk reduction, Ret. risk retention.

Table S1: Risk assessment of all 225 threats generated by medini analyze. Type: C component, P port, L connection (link).

ID	Element	Type	STRIDE	Vec.	S	O	F	P	I	R	Risk	Treat.
T001	Mono + Stereo	C	I	L	1	1	4	10	2.63	11.5	Low	Ret.
T002	Mono + Stereo	C	T	L	10	7	4	1	14.81	60.3	High	Red.
T003	Mono + Stereo	C	D	L	7	10	4	1	11.81	48.3	Med.	Red.
T004	Mono + Stereo	C	S	L	10	7	4	1	14.81	60.3	High	Red.
T005	Mono + Stereo	C	E	L	10	7	4	1	14.81	60.3	High	Red.
T006	Mono + Stereo	C	R	L	1	4	7	1	3.19	13.8	Low	Ret.
T007	Mono + Stereo :: image_out	P	I	L	1	1	4	10	2.63	11.5	Low	Ret.
T008	Mono + Stereo :: image_out	P	T	L	10	7	4	1	14.81	60.3	High	Red.
T009	Mono + Stereo :: image_out	P	D	L	7	10	4	1	11.81	48.3	Med.	Red.
T010	image_out ↔ image_in	L	I	L	1	1	4	10	2.63	11.5	Low	Ret.
T011	image_out ↔ image_in	L	T	L	10	7	4	1	14.81	60.3	High	Red.
T012	image_out ↔ image_in	L	D	L	7	10	4	1	11.81	48.3	Med.	Red.
T013	YOLOv5 fast path	C	I	N	1	1	4	4	2.25	23.5	Med.	Red.
T014	YOLOv5 fast path	C	T	N	7	4	4	1	10.31	104.1	Crit.	Red.
T015	YOLOv5 fast path	C	D	N	4	7	4	1	7.31	74.1	High	Red.
T016	YOLOv5 fast path	C	S	N	7	4	4	1	10.31	104.1	Crit.	Red.
T017	YOLOv5 fast path	C	E	N	7	4	4	1	10.31	104.1	Crit.	Red.
T018	YOLOv5 fast path	C	R	N	1	4	7	1	3.19	32.9	Med.	Red.
T019	YOLOv5 fast path :: image_in	P	I	L	1	1	4	10	2.63	11.5	Low	Ret.
T020	YOLOv5 fast path :: image_in	P	T	L	10	7	4	1	14.81	60.3	High	Red.
T021	YOLOv5 fast path :: image_in	P	D	L	7	10	4	1	11.81	48.3	Med.	Red.
T022	YOLOv5 fast path :: detections_out	P	I	N	1	1	4	4	2.25	23.5	Med.	Red.
T023	YOLOv5 fast path :: detections_out	P	T	N	7	4	4	1	10.31	104.1	Crit.	Red.
T024	YOLOv5 fast path :: detections_out	P	D	N	4	7	4	1	7.31	74.1	High	Red.
T025	detections_out ↔ detections_in	L	I	N	1	1	4	4	2.25	23.5	Med.	Red.
T026	detections_out ↔ detections_in	L	T	N	7	4	4	1	10.31	104.1	Crit.	Red.
T027	detections_out ↔ detections_in	L	D	N	4	7	4	1	7.31	74.1	High	Red.
T028	Agent 1, MLLM call	C	I	N	1	1	4	10	2.63	27.3	Med.	Red.
T029	Agent 1, MLLM call	C	T	N	7	4	4	1	10.31	104.1	Crit.	Red.
T030	Agent 1, MLLM call	C	D	N	4	7	4	1	7.31	74.1	High	Red.
T031	Agent 1, MLLM call	C	S	N	7	4	4	1	10.31	104.1	Crit.	Red.
T032	Agent 1, MLLM call	C	E	N	7	4	4	1	10.31	104.1	Crit.	Red.
T033	Agent 1, MLLM call	C	R	N	1	4	7	1	3.19	32.9	Med.	Red.
T034	Agent 1, MLLM call :: detections_in	P	I	N	1	1	4	10	2.63	27.3	Med.	Red.
T035	Agent 1, MLLM call :: detections_in	P	T	N	7	4	4	1	10.31	104.1	Crit.	Red.
T036	Agent 1, MLLM call :: detections_in	P	D	N	4	7	4	1	7.31	74.1	High	Red.
T037	Agent 1, MLLM call :: hazard_out	P	I	N	1	1	4	10	2.63	27.3	Med.	Red.
T038	Agent 1, MLLM call :: hazard_out	P	T	N	7	4	4	1	10.31	104.1	Crit.	Red.
T039	Agent 1, MLLM call :: hazard_out	P	D	N	4	7	4	1	7.31	74.1	High	Red.
T040	Agent 1, MLLM call :: mllm_request	P	I	N	1	1	4	10	2.63	27.3	Med.	Red.
T041	Agent 1, MLLM call :: mllm_request	P	T	N	7	4	4	1	10.31	104.1	Crit.	Red.
T042	Agent 1, MLLM call :: mllm_request	P	D	N	4	7	4	1	7.31	74.1	High	Red.
T043	Agent 1, MLLM call :: uu_mllm_link	P	I	N	1	1	4	10	2.63	27.3	Med.	Red.
T044	Agent 1, MLLM call :: uu_mllm_link	P	T	N	7	4	4	1	10.31	104.1	Crit.	Red.
T045	Agent 1, MLLM call :: uu_mllm_link	P	D	N	4	7	4	1	7.31	74.1	High	Red.
T046	hazard_out ↔ hazard_in	L	I	N	1	1	4	4	2.25	23.5	Med.	Red.
T047	hazard_out ↔ hazard_in	L	T	N	7	4	4	1	10.31	104.1	Crit.	Red.
T048	hazard_out ↔ hazard_in	L	D	N	4	7	4	1	7.31	74.1	High	Red.
T049	mllm_request ↔ mllm_in	L	I	N	1	1	4	4	2.25	23.5	Med.	Red.
T050	mllm_request ↔ mllm_in	L	T	N	7	4	4	1	10.31	104.1	Crit.	Red.
T051	mllm_request ↔ mllm_in	L	D	N	4	7	4	1	7.31	74.1	High	Red.
T052	uu_mllm_link ↔ mllm_link_1	L	I	N	1	1	4	4	2.25	23.5	Med.	Red.
T053	uu_mllm_link ↔ mllm_link_1	L	T	N	7	4	4	1	10.31	104.1	Crit.	Red.
T054	uu_mllm_link ↔ mllm_link_1	L	D	N	4	7	4	1	7.31	74.1	High	Red.
T055	Agent 2, Depth Pro	C	I	N	1	1	4	4	2.25	23.5	Med.	Red.
T056	Agent 2, Depth Pro	C	T	N	7	4	4	1	10.31	104.1	Crit.	Red.
T057	Agent 2, Depth Pro	C	D	N	4	7	4	1	7.31	74.1	High	Red.
T058	Agent 2, Depth Pro	C	S	N	7	4	4	1	10.31	104.1	Crit.	Red.
T059	Agent 2, Depth Pro	C	E	N	7	4	4	1	10.31	104.1	Crit.	Red.
T060	Agent 2, Depth Pro	C	R	N	1	4	7	1	3.19	32.9	Med.	Red.
T061	Agent 2, Depth Pro :: hazard_in	P	I	N	1	1	4	4	2.25	23.5	Med.	Red.

ID	Element	Type	STRIDE	Vec.	S	O	F	P	I	R	Risk	Treat.
T062	Agent 2, Depth Pro :: hazard_in	P	T	N	7	4	4	1	10.31	104.1	Crit.	Red.
T063	Agent 2, Depth Pro :: hazard_in	P	D	N	4	7	4	1	7.31	74.1	High	Red.
T064	Agent 2, Depth Pro :: distance_out	P	I	N	1	1	4	4	2.25	23.5	Med.	Red.
T065	Agent 2, Depth Pro :: distance_out	P	T	N	7	4	4	1	10.31	104.1	Crit.	Red.
T066	Agent 2, Depth Pro :: distance_out	P	D	N	4	7	4	1	7.31	74.1	High	Red.
T067	distance_out ↔ distance_in	L	I	N	1	1	4	4	2.25	23.5	Med.	Red.
T068	distance_out ↔ distance_in	L	T	N	7	4	4	1	10.31	104.1	Crit.	Red.
T069	distance_out ↔ distance_in	L	D	N	4	7	4	1	7.31	74.1	High	Red.
T070	Agent 3, MLLM call	C	I	N	1	1	4	10	2.63	27.3	Med.	Red.
T071	Agent 3, MLLM call	C	T	N	7	4	4	1	10.31	104.1	Crit.	Red.
T072	Agent 3, MLLM call	C	D	N	4	7	4	1	7.31	74.1	High	Red.
T073	Agent 3, MLLM call	C	S	N	7	4	4	1	10.31	104.1	Crit.	Red.
T074	Agent 3, MLLM call	C	E	N	7	4	4	1	10.31	104.1	Crit.	Red.
T075	Agent 3, MLLM call	C	R	N	1	4	7	1	3.19	32.9	Med.	Red.
T076	Agent 3, MLLM call :: distance_in	P	I	N	1	1	4	10	2.63	27.3	Med.	Red.
T077	Agent 3, MLLM call :: distance_in	P	T	N	7	4	4	1	10.31	104.1	Crit.	Red.
T078	Agent 3, MLLM call :: distance_in	P	D	N	4	7	4	1	7.31	74.1	High	Red.
T079	Agent 3, MLLM call :: content_out	P	I	N	1	1	4	10	2.63	27.3	Med.	Red.
T080	Agent 3, MLLM call :: content_out	P	T	N	7	4	4	1	10.31	104.1	Crit.	Red.
T081	Agent 3, MLLM call :: content_out	P	D	N	4	7	4	1	7.31	74.1	High	Red.
T082	Agent 3, MLLM call :: mllm_request	P	I	N	1	1	4	10	2.63	27.3	Med.	Red.
T083	Agent 3, MLLM call :: mllm_request	P	T	N	7	4	4	1	10.31	104.1	Crit.	Red.
T084	Agent 3, MLLM call :: mllm_request	P	D	N	4	7	4	1	7.31	74.1	High	Red.
T085	Agent 3, MLLM call :: uu_mllm_link	P	I	N	1	1	4	10	2.63	27.3	Med.	Red.
T086	Agent 3, MLLM call :: uu_mllm_link	P	T	N	7	4	4	1	10.31	104.1	Crit.	Red.
T087	Agent 3, MLLM call :: uu_mllm_link	P	D	N	4	7	4	1	7.31	74.1	High	Red.
T088	content_out ↔ content_in	L	I	N	1	1	4	4	2.25	23.5	Med.	Red.
T089	content_out ↔ content_in	L	T	N	7	4	4	1	10.31	104.1	Crit.	Red.
T090	content_out ↔ content_in	L	D	N	4	7	4	1	7.31	74.1	High	Red.
T091	mllm_request ↔ mllm_in_2	L	I	N	1	1	4	4	2.25	23.5	Med.	Red.
T092	mllm_request ↔ mllm_in_2	L	T	N	7	4	4	1	10.31	104.1	Crit.	Red.
T093	mllm_request ↔ mllm_in_2	L	D	N	4	7	4	1	7.31	74.1	High	Red.
T094	uu_mllm_link ↔ mllm_link_2	L	I	N	1	1	4	4	2.25	23.5	Med.	Red.
T095	uu_mllm_link ↔ mllm_link_2	L	T	N	7	4	4	1	10.31	104.1	Crit.	Red.
T096	uu_mllm_link ↔ mllm_link_2	L	D	N	4	7	4	1	7.31	74.1	High	Red.
T097	ASN.1 UPER payload	C	I	A	1	1	4	4	2.25	16.8	Med.	Red.
T098	ASN.1 UPER payload	C	T	A	10	7	4	1	14.81	104.7	Crit.	Red.
T099	ASN.1 UPER payload	C	D	A	4	7	4	1	7.31	52.2	High	Red.
T100	ASN.1 UPER payload	C	S	A	10	7	4	1	14.81	104.7	Crit.	Red.
T101	ASN.1 UPER payload	C	E	A	10	7	4	1	14.81	104.7	Crit.	Red.
T102	ASN.1 UPER payload	C	R	A	1	4	7	1	3.19	23.3	Med.	Red.
T103	ASN.1 UPER payload :: content_in	P	I	A	1	1	4	4	2.25	16.8	Med.	Red.
T104	ASN.1 UPER payload :: content_in	P	T	A	10	7	4	1	14.81	104.7	Crit.	Red.
T105	ASN.1 UPER payload :: content_in	P	D	A	4	7	4	1	7.31	52.2	High	Red.
T106	ASN.1 UPER payload :: denm_out	P	I	A	1	1	4	4	2.25	16.8	Med.	Red.
T107	ASN.1 UPER payload :: denm_out	P	T	A	10	7	4	1	14.81	104.7	Crit.	Red.
T108	ASN.1 UPER payload :: denm_out	P	D	A	4	7	4	1	7.31	52.2	High	Red.
T109	denm_out ↔ denm_in	L	I	A	1	1	4	4	2.25	16.8	Med.	Red.
T110	denm_out ↔ denm_in	L	T	A	10	7	4	1	14.81	104.7	Crit.	Red.
T111	denm_out ↔ denm_in	L	D	A	4	7	4	1	7.31	52.2	High	Red.
T112	Signs, PC5, HSM	C	I	A	1	1	4	4	2.25	16.8	Med.	Red.
T113	Signs, PC5, HSM	C	T	A	10	7	4	1	14.81	104.7	Crit.	Red.
T114	Signs, PC5, HSM	C	D	A	4	7	4	1	7.31	52.2	High	Red.
T115	Signs, PC5, HSM	C	S	A	10	7	4	1	14.81	104.7	Crit.	Red.
T116	Signs, PC5, HSM	C	E	A	10	7	4	1	14.81	104.7	Crit.	Red.
T117	Signs, PC5, HSM	C	R	A	1	4	7	1	3.19	23.3	Med.	Red.
T118	Signs, PC5, HSM :: denm_in	P	I	A	1	1	4	4	2.25	16.8	Med.	Red.
T119	Signs, PC5, HSM :: denm_in	P	T	A	10	7	4	1	14.81	104.7	Crit.	Red.
T120	Signs, PC5, HSM :: denm_in	P	D	A	4	7	4	1	7.31	52.2	High	Red.
T121	Signs, PC5, HSM :: pc5_out	P	I	A	1	1	4	4	2.25	16.8	Med.	Red.
T122	Signs, PC5, HSM :: pc5_out	P	T	A	10	7	4	1	14.81	104.7	Crit.	Red.
T123	Signs, PC5, HSM :: pc5_out	P	D	A	4	7	4	1	7.31	52.2	High	Red.
T124	Signs, PC5, HSM :: uu_uplink	P	I	A	1	1	4	4	2.25	16.8	Med.	Red.
T125	Signs, PC5, HSM :: uu_uplink	P	T	A	10	7	4	1	14.81	104.7	Crit.	Red.
T126	Signs, PC5, HSM :: uu_uplink	P	D	A	4	7	4	1	7.31	52.2	High	Red.
T127	Signs, PC5, HSM :: uu_link	P	I	A	1	1	4	4	2.25	16.8	Med.	Red.
T128	Signs, PC5, HSM :: uu_link	P	T	A	10	7	4	1	14.81	104.7	Crit.	Red.
T129	Signs, PC5, HSM :: uu_link	P	D	A	4	7	4	1	7.31	52.2	High	Red.
T130	pc5_out ↔ pc5_in	L	I	N	1	1	4	4	2.25	23.5	Med.	Red.
T131	pc5_out ↔ pc5_in	L	T	N	7	4	4	1	10.31	104.1	Crit.	Red.
T132	pc5_out ↔ pc5_in	L	D	N	4	7	4	1	7.31	74.1	High	Red.
T133	uu_uplink ↔ uu_in	L	I	N	1	1	4	4	2.25	23.5	Med.	Red.
T134	uu_uplink ↔ uu_in	L	T	N	7	4	4	1	10.31	104.1	Crit.	Red.
T135	uu_uplink ↔ uu_in	L	D	N	4	7	4	1	7.31	74.1	High	Red.
T136	uu_link ↔ uu_link_rsu	L	I	N	1	1	4	4	2.25	23.5	Med.	Red.

ID	Element	Type	STRIDE	Vec.	S	O	F	P	I	R	Risk	Treat.
T137	uu_link ↔ uu_link_rsu	L	T	N	7	4	4	1	10.31	104.1	Crit.	Red.
T138	uu_link ↔ uu_link_rsu	L	D	N	4	7	4	1	7.31	74.1	High	Red.
T139	PC5 + Uumodem	C	I	A	1	1	4	4	2.25	16.8	Med.	Red.
T140	PC5 + Uumodem	C	T	A	10	7	4	1	14.81	104.7	Crit.	Red.
T141	PC5 + Uumodem	C	D	A	4	7	4	1	7.31	52.2	High	Red.
T142	PC5 + Uumodem	C	S	A	10	7	4	1	14.81	104.7	Crit.	Red.
T143	PC5 + Uumodem	C	E	A	10	7	4	1	14.81	104.7	Crit.	Red.
T144	PC5 + Uumodem	C	R	A	1	4	7	1	3.19	23.3	Med.	Red.
T145	PC5 + Uumodem :: pc5_in	P	I	A	1	1	4	4	2.25	16.8	Med.	Red.
T146	PC5 + Uumodem :: pc5_in	P	T	A	10	7	4	1	14.81	104.7	Crit.	Red.
T147	PC5 + Uumodem :: pc5_in	P	D	A	4	7	4	1	7.31	52.2	High	Red.
T148	PC5 + Uumodem :: warning_out	P	I	A	1	1	4	4	2.25	16.8	Med.	Red.
T149	PC5 + Uumodem :: warning_out	P	T	A	10	7	4	1	14.81	104.7	Crit.	Red.
T150	PC5 + Uumodem :: warning_out	P	D	A	7	10	4	1	11.81	83.7	High	Red.
T151	PC5 + Uumodem :: uu_modem	P	I	A	1	1	4	4	2.25	16.8	Med.	Red.
T152	PC5 + Uumodem :: uu_modem	P	T	A	10	7	4	1	14.81	104.7	Crit.	Red.
T153	PC5 + Uumodem :: uu_modem	P	D	A	4	7	4	1	7.31	52.2	High	Red.
T154	PC5 + Uumodem :: uu_modem	P	I	A	1	1	4	4	2.25	16.8	Med.	Red.
T155	PC5 + Uumodem :: uu_modem	P	T	A	10	7	4	1	14.81	104.7	Crit.	Red.
T156	PC5 + Uumodem :: uu_modem	P	D	A	4	7	4	1	7.31	52.2	High	Red.
T157	warning_out ↔ warning_in	L	I	L	1	1	4	4	2.25	10.0	Low	Ret.
T158	warning_out ↔ warning_in	L	T	L	10	7	4	1	14.81	60.3	High	Red.
T159	warning_out ↔ warning_in	L	D	L	7	10	4	1	11.81	48.3	Med.	Red.
T160	uu_modem ↔ uu_in_2	L	I	N	1	1	4	4	2.25	23.5	Med.	Red.
T161	uu_modem ↔ uu_in_2	L	T	N	7	4	4	1	10.31	104.1	Crit.	Red.
T162	uu_modem ↔ uu_in_2	L	D	N	4	7	4	1	7.31	74.1	High	Red.
T163	Verify, warning logic	C	I	L	1	1	4	4	2.25	10.0	Low	Ret.
T164	Verify, warning logic	C	T	L	10	7	4	1	14.81	60.3	High	Red.
T165	Verify, warning logic	C	D	L	7	10	4	1	11.81	48.3	Med.	Red.
T166	Verify, warning logic	C	S	L	10	7	4	1	14.81	60.3	High	Red.
T167	Verify, warning logic	C	E	L	10	7	4	1	14.81	60.3	High	Red.
T168	Verify, warning logic	C	R	L	1	4	7	1	3.19	13.8	Low	Ret.
T169	Verify, warning logic :: warning_in	P	I	L	1	1	4	4	2.25	10.0	Low	Ret.
T170	Verify, warning logic :: warning_in	P	T	L	10	7	4	1	14.81	60.3	High	Red.
T171	Verify, warning logic :: warning_in	P	D	L	7	10	4	1	11.81	48.3	Med.	Red.
T172	Verify, warning logic :: alert_out	P	I	L	1	1	4	4	2.25	10.0	Low	Ret.
T173	Verify, warning logic :: alert_out	P	T	L	10	7	4	1	14.81	60.3	High	Red.
T174	Verify, warning logic :: alert_out	P	D	L	7	10	4	1	11.81	48.3	Med.	Red.
T175	alert_out ↔ alert_in	L	I	L	1	1	4	4	2.25	10.0	Low	Ret.
T176	alert_out ↔ alert_in	L	T	L	10	7	4	1	14.81	60.3	High	Red.
T177	alert_out ↔ alert_in	L	D	L	7	10	4	1	11.81	48.3	Med.	Red.
T178	Visual + audio alert	C	I	L	1	1	4	4	2.25	10.0	Low	Ret.
T179	Visual + audio alert	C	T	L	10	7	4	1	14.81	60.3	High	Red.
T180	Visual + audio alert	C	D	L	7	10	4	1	11.81	48.3	Med.	Red.
T181	Visual + audio alert	C	S	L	10	7	4	1	14.81	60.3	High	Red.
T182	Visual + audio alert	C	E	L	10	7	4	1	14.81	60.3	High	Red.
T183	Visual + audio alert	C	R	L	1	4	7	1	3.19	13.8	Low	Ret.
T184	Visual + audio alert :: alert_in	P	I	L	1	1	4	4	2.25	10.0	Low	Ret.
T185	Visual + audio alert :: alert_in	P	T	L	10	7	4	1	14.81	60.3	High	Red.
T186	Visual + audio alert :: alert_in	P	D	L	7	10	4	1	11.81	48.3	Med.	Red.
T187	Gemini API	C	I	N	1	1	4	10	2.63	27.3	Med.	Red.
T188	Gemini API	C	T	N	7	4	4	1	10.31	104.1	Crit.	Red.
T189	Gemini API	C	D	N	4	7	4	1	7.31	74.1	High	Red.
T190	Gemini API	C	S	N	7	4	4	1	10.31	104.1	Crit.	Red.
T191	Gemini API	C	E	N	7	4	4	1	10.31	104.1	Crit.	Red.
T192	Gemini API	C	R	N	1	4	7	1	3.19	32.9	Med.	Red.
T193	Gemini API :: mllm_in	P	I	N	1	1	4	10	2.63	27.3	Med.	Red.
T194	Gemini API :: mllm_in	P	T	N	7	4	4	1	10.31	104.1	Crit.	Red.
T195	Gemini API :: mllm_in	P	D	N	4	7	4	1	7.31	74.1	High	Red.
T196	Gemini API :: mllm_in_2	P	I	N	1	1	4	10	2.63	27.3	Med.	Red.
T197	Gemini API :: mllm_in_2	P	T	N	7	4	4	1	10.31	104.1	Crit.	Red.
T198	Gemini API :: mllm_in_2	P	D	N	4	7	4	1	7.31	74.1	High	Red.
T199	Gemini API :: mllm_link_1	P	I	N	1	1	4	10	2.63	27.3	Med.	Red.
T200	Gemini API :: mllm_link_1	P	T	N	7	4	4	1	10.31	104.1	Crit.	Red.
T201	Gemini API :: mllm_link_1	P	D	N	4	7	4	1	7.31	74.1	High	Red.
T202	Gemini API :: mllm_link_2	P	I	N	1	1	4	10	2.63	27.3	Med.	Red.
T203	Gemini API :: mllm_link_2	P	T	N	7	4	4	1	10.31	104.1	Crit.	Red.
T204	Gemini API :: mllm_link_2	P	D	N	4	7	4	1	7.31	74.1	High	Red.
T205	PKI, operator	C	I	A	1	1	4	4	2.25	16.8	Med.	Red.
T206	PKI, operator	C	T	A	10	7	4	1	14.81	104.7	Crit.	Red.
T207	PKI, operator	C	D	A	4	7	4	1	7.31	52.2	High	Red.
T208	PKI, operator	C	S	A	10	7	4	1	14.81	104.7	Crit.	Red.
T209	PKI, operator	C	E	A	10	7	4	1	14.81	104.7	Crit.	Red.
T210	PKI, operator	C	R	A	1	4	7	1	3.19	23.3	Med.	Red.
T211	PKI, operator :: uu_in	P	I	A	1	1	4	4	2.25	16.8	Med.	Red.

ID	Element	Type	STRIDE	Vec.	<i>S</i>	<i>O</i>	<i>F</i>	<i>P</i>	<i>I</i>	<i>R</i>	Risk	Treat.
T212	PKI, operator :: uu_in	P	T	A	10	7	4	1	14.81	104.7	Crit.	Red.
T213	PKI, operator :: uu_in	P	D	A	4	7	4	1	7.31	52.2	High	Red.
T214	PKI, operator :: uu_in_2	P	I	A	1	1	4	4	2.25	16.8	Med.	Red.
T215	PKI, operator :: uu_in_2	P	T	A	10	7	4	1	14.81	104.7	Crit.	Red.
T216	PKI, operator :: uu_in_2	P	D	A	4	7	4	1	7.31	52.2	High	Red.
T217	PKI, operator :: uu_link_rsu	P	I	A	1	1	4	4	2.25	16.8	Med.	Red.
T218	PKI, operator :: uu_link_rsu	P	T	A	10	7	4	1	14.81	104.7	Crit.	Red.
T219	PKI, operator :: uu_link_rsu	P	D	A	4	7	4	1	7.31	52.2	High	Red.
T220	PKI, operator :: uu_modem_out	P	I	A	1	1	4	4	2.25	16.8	Med.	Red.
T221	PKI, operator :: uu_modem_out	P	T	A	10	7	4	1	14.81	104.7	Crit.	Red.
T222	PKI, operator :: uu_modem_out	P	D	A	4	7	4	1	7.31	52.2	High	Red.
T223	uu_modem_out ↔ uu_modem	L	I	N	1	1	4	4	2.25	23.5	Med.	Red.
T224	uu_modem_out ↔ uu_modem	L	T	N	7	4	4	1	10.31	104.1	Crit.	Red.
T225	uu_modem_out ↔ uu_modem	L	D	N	4	7	4	1	7.31	74.1	High	Red.

Table S2: Risk assessment of all 170 LLM threat scenarios. Row gives the row in the LLM worksheet; Element gives the architecture element used for matching; Medini gives the identifier of the medini component threat with the same element and STRIDE category, “link” where the pair is enumerated on connections only, and – where medini does not enumerate the pair.

Row	Model	Element	ST.	Vec.	Scenario	R	Risk	Medini
2	ChatGPT	Cameras	S	L	An attacker injects a forged camera stream or replays previously recorded road images so that the accident-detection system perceives a false traffic situation.	60.3	High	T004
3	ChatGPT	Cameras	T	L	Camera firmware, calibration parameters, timestamps, or image frames are modified in transit or at the roadside ITS station, causing incorrect object detection or distance estimation.	60.3	High	T002
4	ChatGPT	Cameras	I	L	Unauthorized access to live or stored camera imagery exposes vehicle occupants, license plates, road users, or other potentially sensitive visual information.	11.5	Low	T001
5	ChatGPT	Cameras	D	L	An attacker jams, floods, disconnects, or disables the camera interface, preventing the system from detecting accidents.	48.3	Med.	T003
6	ChatGPT	Edge detector	S	L	Adversarial or fabricated images are supplied to the detector so that nonexistent vehicles, pedestrians, or hazards are detected.	60.3	High	T016
7	ChatGPT	Edge detector	T	L	The YOLOv5 model, weights, configuration, preprocessing pipeline, or confidence thresholds are modified to suppress or create detections.	60.3	High	T014
8	ChatGPT	Edge detector	I	L	Intermediate detections, image buffers, or debug interfaces are exposed to an unauthorized party.	11.5	Low	T013
9	ChatGPT	Edge detector	D	L	Malformed inputs or excessive image frames exhaust CPU/GPU resources and prevent real-time detection.	48.3	Med.	T015
10	ChatGPT	Edge detector	E	L	A vulnerability in the detector runtime or inference service is exploited to obtain operating-system privileges on the roadside station.	60.3	High	T017
11	ChatGPT	Agent 1	S	N	An attacker impersonates the trusted MLLM service or injects fabricated scene descriptions into the agent’s input.	104.1	Crit.	T031
12	ChatGPT	Agent 1	T	N	Prompts, agent context, system instructions, tool parameters, or detected-object data are modified so the agent incorrectly classifies a non-accident as an accident or misses a real accident.	104.1	Crit.	T029
13	ChatGPT	Agent 1	I	N	Camera-derived scene descriptions, location information, or accident context are disclosed to an unauthorized cloud service, API consumer, or attacker.	27.3	Med.	T028
14	ChatGPT	Agent 1	D	N	Repeated or oversized MLLM requests consume API quotas or network resources and delay situation classification.	74.1	High	T030
15	ChatGPT	Agent 1	E	N	Prompt injection or agent-service vulnerabilities cause the agent to invoke unauthorized tools, APIs, or local system functions.	104.1	Crit.	T032
16	ChatGPT	Cloud MLLM	S	N	A rogue endpoint impersonates the cloud MLLM API, causing the roadside system to send accident data to an attacker-controlled service.	104.1	Crit.	T190
17	ChatGPT	Cloud MLLM	T	N	Requests or responses between the roadside station and cloud MLLM are modified, potentially changing accident classification or generated content.	104.1	Crit.	T188
18	ChatGPT	Cloud MLLM	R	N	A lack of reliable request, response, and identity logging makes it difficult to determine which API account generated a critical accident-analysis result.	32.9	Med.	T192
19	ChatGPT	Cloud MLLM	I	N	Images, scene descriptions, location, or vehicle-related information sent to the cloud are exposed through compromised credentials, insecure API configuration, or unauthorized access.	27.3	Med.	T187
20	ChatGPT	Cloud MLLM	D	N	API quota exhaustion, credential abuse, network disruption, or provider unavailability prevents timely MLLM analysis.	74.1	High	T189
21	ChatGPT	Cloud MLLM	E	N	Compromised API credentials are reused to obtain access to additional cloud resources, data, or administrative functions.	104.1	Crit.	T191
22	ChatGPT	Agent 2	S	L	A manipulated stereo/depth input causes the component to estimate a false distance to a vehicle, pedestrian, or obstacle.	60.3	High	T058
23	ChatGPT	Agent 2	T	L	Depth-model weights, calibration data, or processing parameters are altered so that estimated distances are systematically incorrect.	60.3	High	T056
24	ChatGPT	Agent 2	I	L	Depth maps or intermediate spatial data are exposed, revealing the structure of scenes captured by roadside cameras.	11.5	Low	T055
25	ChatGPT	Agent 2	D	L	Excessive inference requests or malformed depth inputs cause the distance-estimation service to become unavailable.	48.3	Med.	T057
26	ChatGPT	Agent 3	S	N	An attacker impersonates the message-generation service and generates a fraudulent safety warning or DENM message.	104.1	Crit.	T073
27	ChatGPT	Agent 3	T	N	Accident context, distance information, warning text, severity, location, or message parameters are changed before DENM encoding.	104.1	Crit.	T071
28	ChatGPT	Agent 3	R	N	Insufficient provenance information prevents tracing a transmitted warning back to the specific agent, input event, and model response.	32.9	Med.	T075
29	ChatGPT	Agent 3	I	N	The generated message or its underlying accident context reveals sensitive location or vehicle information.	27.3	Med.	T070
30	ChatGPT	Agent 3	D	N	The message-generation agent is overwhelmed, causing safety messages to be delayed beyond their useful lifetime.	74.1	High	T072
31	ChatGPT	Agent 3	E	N	Prompt injection or compromised agent credentials allow unauthorized access to message-generation APIs or downstream ITS functions.	104.1	Crit.	T074
32	ChatGPT	DENM encoder	S	A	An attacker constructs a valid-looking DENM payload and injects it into the communication path while pretending to be the roadside ITS station.	104.7	Crit.	T100
33	ChatGPT	DENM encoder	T	A	DENM fields such as event type, coordinates, severity, timestamp, validity period, or relevance area are modified before transmission.	104.7	Crit.	T098

Row	Model	Element	ST.	Vec.	Scenario	R	Risk	Medini
34	ChatGPT	DENM encoder	R	A	Without end-to-end message identifiers and audit records, the sender may later deny having generated a particular DENM event.	23.3	Med.	T102
35	ChatGPT	DENM encoder	I	A	Unprotected DENM payloads expose accident location, event details, timestamps, or other contextual information to passive observers.	16.8	Med.	T097
36	ChatGPT	DENM encoder	D	A	Malformed ASN.1/UPER payloads trigger parser failures, excessive processing, or rejection of legitimate safety messages.	52.2	High	T099
37	ChatGPT	RSU	S	A	An attacker impersonates the RSU and broadcasts forged accident or road-hazard notifications to nearby vehicles.	104.7	Crit.	T115
38	ChatGPT	RSU	T	A	RSU configuration, firmware, signed-message handling, or radio parameters are modified to alter transmitted safety information.	104.7	Crit.	T113
39	ChatGPT	RSU	R	A	Insufficient security logging prevents attribution of signed or transmitted messages to a particular RSU instance or administrative operator.	23.3	Med.	T117
40	ChatGPT	RSU	I	A	RSU management interfaces, PC5 traffic, or stored event information are accessed by an unauthorized party.	16.8	Med.	T112
41	ChatGPT	RSU	D	A	RF jamming, flooding, repeated invalid messages, or exploitation of RSU services prevents timely safety-message broadcasting.	52.2	High	T114
42	ChatGPT	RSU	E	A	A compromised RSU service account is used to gain administrative control over signing functions, radio configuration, or HSM operations.	104.7	Crit.	T116
43	ChatGPT	Central ITS-S	S	A	An attacker steals or forges ITS credentials or certificates and impersonates a legitimate RSU, vehicle, or backend identity.	104.7	Crit.	T208
44	ChatGPT	Central ITS-S	T	A	Certificate status, trust relationships, revocation information, or operator-controlled PKI configuration are modified.	104.7	Crit.	T206
45	ChatGPT	Central ITS-S	R	A	Weak PKI audit trails make it difficult to prove which operator or process issued, revoked, or renewed a credential.	23.3	Med.	T210
46	ChatGPT	Central ITS-S	I	A	Private keys, certificate metadata, enrollment information, or operator records are disclosed to unauthorized parties.	19.4	Med.	T205
47	ChatGPT	Central ITS-S	D	A	The PKI, certificate-status service, or enrollment service is made unavailable, causing legitimate ITS entities to lose the ability to authenticate or validate messages.	52.2	High	T207
48	ChatGPT	Central ITS-S	E	A	Compromise of a PKI administrator account allows an attacker to issue trusted certificates for unauthorized systems.	104.7	Crit.	T209
49	ChatGPT	Certificates and reports	S	N	Forged certificates or fabricated event reports are presented as being generated by legitimate ITS entities.	104.1	Crit.	–
50	ChatGPT	Certificates and reports	T	N	Event reports are modified after generation, including accident time, location, severity, sensor evidence, or responsible system identity.	104.1	Crit.	link
51	ChatGPT	Certificates and reports	R	N	Missing timestamps, signatures, hash chains, or audit records prevent forensic attribution of accident reports and certificate operations.	32.9	Med.	–
52	ChatGPT	Certificates and reports	I	N	Event reports expose vehicle identifiers, road locations, camera-derived information, or operational metadata to unauthorized users.	23.5	Med.	link
53	ChatGPT	Certificates and reports	D	N	Attackers delete, corrupt, or flood the event-reporting store, preventing investigation and operational monitoring.	74.1	High	link
54	ChatGPT	OBU	S	A	A rogue device impersonates a legitimate vehicle OBU or roadside node and sends forged safety messages.	104.7	Crit.	T142
55	ChatGPT	OBU	T	A	PC5 or Uu messages are modified, replayed, or reordered so that the vehicle receives an incorrect accident warning.	104.7	Crit.	T140
56	ChatGPT	OBU	R	A	Inadequate message-level evidence prevents proving which OBU transmitted or received a particular safety event.	23.3	Med.	T144
57	ChatGPT	OBU	I	A	Unauthorized listeners infer vehicle identity, trajectory, location, or nearby accident information from wireless traffic.	16.8	Med.	T139
58	ChatGPT	OBU	D	A	PC5/Uu jamming, signaling floods, malformed packets, or modem resource exhaustion prevent reception of urgent safety messages.	52.2	High	T141
59	ChatGPT	OBU	E	A	A modem or OBU vulnerability is exploited to gain privileged access to vehicle-network interfaces.	104.7	Crit.	T143
60	ChatGPT	Vehicle computer	S	L	Forged but apparently valid messages are presented to the vehicle computer as originating from a trusted RSU or authorized ITS entity.	60.3	High	T166
61	ChatGPT	Vehicle computer	T	L	Verification logic, certificate validation, warning thresholds, software binaries, or configuration parameters are modified to suppress or alter alerts.	60.3	High	T164
62	ChatGPT	Vehicle computer	R	L	Lack of secure logs prevents forensic determination of which warning was received, verified, rejected, or presented.	13.8	Low	T168
63	ChatGPT	Vehicle computer	I	L	Vehicle computer logs or message buffers expose trip, location, accident, or communication information.	11.5	Low	T163
64	ChatGPT	Vehicle computer	D	L	A flood of invalid or computationally expensive messages exhausts verification resources and delays processing of genuine safety events.	48.3	Med.	T165
65	ChatGPT	Vehicle computer	E	L	Exploitation of the warning-processing software provides unauthorized access to privileged vehicle-network functions.	60.3	High	T167
66	ChatGPT	Driver HMI	S	L	A malicious component generates a false safety alert that appears to originate from the trusted accident-detection system.	60.3	High	T181
67	ChatGPT	Driver HMI	T	L	Alert severity, visual content, audio content, timing, or suppression settings are modified so the driver does not receive the correct warning.	60.3	High	T179
68	ChatGPT	Driver HMI	R	L	The system cannot establish whether a warning was actually displayed or audibly presented to the driver because HMI events were not securely logged.	13.8	Low	T183
69	ChatGPT	Driver HMI	I	L	HMI diagnostic interfaces expose accident notifications, vehicle location, or system-state information to unauthorized users.	11.5	Low	T178
70	ChatGPT	Driver HMI	D	L	The display/audio subsystem is disabled, overloaded, or flooded with alerts, causing a genuine warning to be missed.	48.3	Med.	T180

Row	Model	Element	ST.	Vec.	Scenario	R	Risk	Medini
71	ChatGPT	Driver HMI	E	L	A compromised HMI interface is used as a path to obtain higher privileges on the vehicle computer or other in-vehicle systems.	60.3	High	T182
72	ChatGPT	5G Uu links	S	N	A malicious endpoint impersonates the intended cloud or roadside service using stolen credentials or compromised network identities.	104.1	Crit.	–
73	ChatGPT	5G Uu links	T	N	Requests, MLLM responses, certificates, or event data are altered while traversing the cellular network.	104.1	Crit.	link
74	ChatGPT	5G Uu links	I	N	Unprotected or improperly authenticated traffic reveals accident imagery, event metadata, location, or API credentials.	27.3	Med.	link
75	ChatGPT	5G Uu links	D	N	Cellular jamming, signaling attacks, traffic flooding, or carrier-side disruption prevents cloud interaction during an accident.	74.1	High	link
76	ChatGPT	PC5 sidelink	S	A	A malicious transmitter sends forged vehicle or RSU safety messages over PC5.	104.7	Crit.	–
77	ChatGPT	PC5 sidelink	T	A	Wireless messages are altered, replayed, delayed, or reordered so that vehicles receive inaccurate accident information.	104.7	Crit.	link
78	ChatGPT	PC5 sidelink	I	A	Passive radio monitoring reveals safety-event details, device identifiers, or movement-related information.	16.8	Med.	link
79	ChatGPT	PC5 sidelink	D	A	RF interference, protocol flooding, or repeated malformed packets prevent delivery of legitimate PC5 safety messages.	52.2	High	link
80	ChatGPT	AI models and prompts	S	N	An unauthorized model or agent presents itself as an approved accident-detection component.	104.1	Crit.	–
81	ChatGPT	AI models and prompts	T	N	Model files, prompts, system instructions, agent policies, safety rules, or threshold parameters are altered to induce unsafe decisions.	104.1	Crit.	–
82	ChatGPT	AI models and prompts	R	N	Absence of model/version and prompt provenance makes it impossible to determine which AI configuration produced a safety-critical decision.	32.9	Med.	–
83	ChatGPT	AI models and prompts	I	N	Proprietary prompts, model outputs, inference data, or training-related information are exposed through debugging or service interfaces.	23.5	Med.	–
84	ChatGPT	AI models and prompts	D	N	Resource-intensive or adversarial inputs force AI inference components beyond their real-time processing capacity.	74.1	High	–
85	ChatGPT	AI models and prompts	E	N	An AI agent is manipulated through prompt injection into accessing tools, files, APIs, or operating-system resources outside its intended authority.	104.1	Crit.	–
86	ChatGPT	End-to-end data flow	S	N	An attacker inserts a false accident event into the pipeline so that downstream components treat it as a trusted detection.	104.1	Crit.	–
87	ChatGPT	End-to-end data flow	T	N	Event state, timestamps, location, distance, severity, or identity are modified between detection, AI analysis, DENM generation, RSU transmission, and vehicle verification.	104.1	Crit.	–
88	ChatGPT	End-to-end data flow	R	N	Missing end-to-end correlation identifiers, signatures, synchronized timestamps, and audit logs prevent reconstruction of the complete accident-detection chain.	32.9	Med.	–
89	ChatGPT	End-to-end data flow	I	N	Sensitive event data is exposed at intermediate processing stages, including camera images, AI prompts, model responses, and vehicle communication metadata.	23.5	Med.	–
90	ChatGPT	End-to-end data flow	D	N	An attacker disrupts one or more pipeline stages so that a valid accident is not detected, analyzed, transmitted, verified, or presented to the driver in time.	74.1	High	–
91	ChatGPT	End-to-end data flow	E	N	Compromise of a low-trust component, such as a camera or exposed API, is used as a pivot to reach privileged AI, PKI, RSU, OBU, or vehicle-control functions.	104.1	Crit.	–
92	Gemini	Cameras	S	L	An attacker feeds a pre-recorded or manipulated video stream into the camera interface to simulate a false traffic accident.	60.3	High	T004
93	Gemini	Cameras	T	L	Physically altering, blinding with lasers, or damaging the camera sensors to disrupt visual input to the Roadside ITS station.	60.3	High	T002
94	Gemini	Cameras	D	L	Physically covering the camera lens or overwhelming the sensor with intense light to prevent image capture.	48.3	Med.	T003
95	Gemini	Edge detector	T	L	Adversarial machine learning attacks via slight physical alterations to the environment causing YOLOv5 or the MLLM agents to misclassify objects, accident severity, or distance estimates.	60.3	High	T014
96	Gemini	Edge detector	E	L	Exploiting software vulnerabilities in the local AI agents or underlying OS to execute arbitrary code and gain unauthorized control over the Roadside ITS station.	60.3	High	T017
97	Gemini	Cloud MLLM	I	N	Interception of unencrypted or weakly encrypted 5G traffic, exposing sensitive visual data sent to the Cloud MLLM.	27.3	Med.	T187
98	Gemini	Cloud MLLM	D	N	RF jamming of the 5G Uu cellular link, preventing the Roadside ITS station from communicating with the Cloud MLLM for situation detection and message generation.	74.1	High	T189
99	Gemini	Cloud MLLM	T	N	Man-in-the-Middle (MitM) attacks altering the prompt sent to the Gemini API or the returning hazard classification, assuming TLS is bypassed or improperly validated.	104.1	Crit.	T188
100	Gemini	DENM encoder	T	A	Exploiting parsing vulnerabilities such as buffer overflows in the ASN.1 UPER encoder to crash the RSU or execute arbitrary code.	104.7	Crit.	T098
101	Gemini	DENM encoder	S	A	A compromised device impersonates the legitimate RSU by extracting keys from a vulnerable HSM to broadcast fake hazard warnings.	104.7	Crit.	T100
102	Gemini	DENM encoder	I	A	Extracting private signing keys or certificates stored improperly within the RSU's Hardware Security Module (HSM).	16.8	Med.	T097
103	Gemini	Central ITS-S	S	A	Compromise of the Root or Subordinate CA within the PKI, allowing an attacker to issue fraudulent certificates for rogue RSUs or OBUs.	104.7	Crit.	T208
104	Gemini	Central ITS-S	E	A	An attacker gains unauthorized access to the operator console, allowing them to revoke legitimate certificates or issue false network-wide event reports.	104.7	Crit.	T209

Row	Model	Element	ST.	Vec.	Scenario	R	Risk	Medini
105	Gemini	Central ITS-S	R	A	Insufficient logging mechanisms at the Central ITS-S allow a malicious insider or compromised node to delete traces of fraudulent certificate issuance or false event reporting.	23.3	Med.	T210
106	Gemini	OBU	D	A	RF jamming of the PC5 sidelink frequency, preventing the OBU from receiving critical DENM alerts from the RSU.	52.2	High	T141
107	Gemini	OBU	S	A	Sybil attack where an attacker broadcasts false vehicle telemetry over the PC5 interface using multiple forged OBU identities to trigger false collision warnings.	104.7	Crit.	T142
108	Gemini	Vehicle computer	T	L	Malware infection on the vehicle computer altering the warning logic to suppress legitimate collision alerts or trigger continuous false alarms on the Driver HMI.	60.3	High	T164
109	Gemini	Vehicle computer	I	L	Exploiting the OBU-to-Vehicle computer interface to extract sensitive vehicle telemetry and driver behavior data.	11.5	Low	T163
110	Claude	Cameras	S	L	Attacker replaces or spoofs the camera feed (e.g., video injection) to feed fabricated scenes into the pipeline.	60.3	High	T004
111	Claude	Cameras	T	L	Physical tampering with camera lens/position, or manipulation of the raw video stream in transit to the edge detector.	60.3	High	T002
112	Claude	Cameras	D	L	Camera blinded/jammed (laser, IR, physical obstruction) or feed flooded to prevent hazard detection.	48.3	Med.	T003
113	Claude	Cameras	I	L	Unauthorized interception of raw video stream exposing personal data (license plates, pedestrian faces).	11.5	Low	T001
114	Claude	Edge detector	T	L	Adversarial perturbation/patch attacks on input images causing misclassification of hazards (evasion attack).	60.3	High	T014
115	Claude	Edge detector	E	L	Exploiting a vulnerability in the inference runtime to gain code execution on the roadside edge device.	60.3	High	T017
116	Claude	Edge detector	D	L	Resource exhaustion via crafted high-complexity frames causing inference latency spikes or crashes.	48.3	Med.	T015
117	Claude	Edge detector	R	L	Lack of logging/integrity on model version used, allowing denial that a faulty/outdated model produced a missed detection.	13.8	Low	T018
118	Claude	Agent 1	T	N	Prompt injection via manipulated image/metadata causing the MLLM call to misclassify or ignore a real hazard.	104.1	Crit.	T029
119	Claude	Agent 1	S	N	Malicious actor spoofs the agent's request to the cloud MLLM, injecting a fabricated scene description.	104.1	Crit.	T031
120	Claude	Agent 1	D	N	Flooding Agent 1 with requests to exhaust cloud MLLM API quota/rate limits, delaying real hazard detection.	74.1	High	T030
121	Claude	Agent 1	I	N	Sensitive roadside imagery sent to external Cloud MLLM (Gemini API) is exposed via API logging or third-party breach.	23.5	Med.	T028
122	Claude	Cloud MLLM	S	N	Man-in-the-middle attack impersonating the Cloud MLLM endpoint to return falsified hazard/message data.	104.1	Crit.	T190
123	Claude	Cloud MLLM	T	N	API response tampered in transit, altering severity classification or generated DENM message content.	104.1	Crit.	T188
124	Claude	Cloud MLLM	D	N	Cloud API outage, throttling, or targeted DDoS on the API endpoint disrupts situation detection and message generation.	74.1	High	T189
125	Claude	Cloud MLLM	I	N	Third-party cloud provider retains/logs sensitive roadside imagery or location data beyond intended use.	27.3	Med.	T187
126	Claude	Cloud MLLM	E	N	Compromised API credentials/key allow attacker to submit arbitrary requests or exfiltrate historical query data.	104.1	Crit.	T191
127	Claude	Agent 2	T	L	Manipulation of depth estimation inputs/outputs to under- or over-report hazard distance, causing incorrect warning timing.	60.3	High	T056
128	Claude	Agent 2	D	L	Malformed stereo input causes depth model to hang or crash, stalling the pipeline before message generation.	48.3	Med.	T057
129	Claude	Agent 3	T	N	Injection attack alters generated hazard message content/severity before DENM encoding.	104.1	Crit.	T071
130	Claude	Agent 3	R	N	No cryptographic binding between agent output and final DENM payload, allowing denial of message origin/authorship.	32.9	Med.	T075
131	Claude	Agent 3	D	N	Malformed or oversized MLLM output causes downstream DENM encoder to fail or drop messages.	74.1	High	T072
132	Claude	DENM encoder	T	A	Malformed or crafted input triggers ASN.1 parser vulnerabilities (buffer overflow) enabling payload corruption or code execution.	104.7	Crit.	T098
133	Claude	DENM encoder	S	A	Injection of a fraudulent DENM message without going through legitimate detection pipeline (bypassing Agents 1-3).	104.7	Crit.	T100
134	Claude	DENM encoder	R	A	Absence of signed audit trail linking each DENM message to the specific detection event that generated it.	23.3	Med.	T102
135	Claude	RSU	S	A	Rogue RSU impersonation broadcasting false DENM messages to nearby vehicles (fake hazard alerts).	104.7	Crit.	T115
136	Claude	RSU	T	A	Physical/firmware tampering with the RSU to alter broadcast messages or extract HSM key material.	104.7	Crit.	T113
137	Claude	RSU	D	A	RF jamming of PC5 sidelink or flooding RSU with bogus messages, preventing legitimate DENM broadcast.	52.2	High	T114
138	Claude	RSU	E	A	Exploitation of RSU management interface to gain administrative control and reconfigure signage/broadcast behavior.	104.7	Crit.	T116
139	Claude	RSU	I	A	Side-channel or physical attack on HSM leaks private signing keys used for message authentication.	16.8	Med.	T112
140	Claude	Central ITS-S	S	A	Attacker impersonates the Central ITS-S to issue fraudulent certificates to rogue RSUs/OBUs.	104.7	Crit.	T208
141	Claude	Central ITS-S	T	A	Compromise of the PKI root/intermediate CA allowing forged or revoked certificates to remain trusted.	104.7	Crit.	T206
142	Claude	Central ITS-S	E	A	Insider or compromised operator account used to issue unauthorized certificates or push malicious configuration.	104.7	Crit.	T209

Row	Model	Element	ST.	Vec.	Scenario	R	Risk	Medini
143	Claude	Central ITS-S	D	A	Attack on certificate revocation/issuance services disrupts trust chain validation across the fleet.	52.2	High	T207
144	Claude	Central ITS-S	R	A	Insufficient logging of certificate issuance/revocation events prevents accountability for a rogue trust decision.	23.3	Med.	T210
145	Claude	OBU	S	A	Attacker spoofs a legitimate RSU/vehicle identity over PC5 to inject false DENM messages directly to the OBU.	104.7	Crit.	T142
146	Claude	OBU	T	A	Firmware tampering or malware installed on the OBU to alter received message handling or relay falsified data.	104.7	Crit.	T140
147	Claude	OBU	D	A	Jamming of PC5/Uu radio interfaces or message flooding overwhelms OBU processing, dropping legitimate alerts.	52.2	High	T141
148	Claude	OBU	I	A	Eavesdropping on Uu cellular link exposes vehicle location and event report data.	16.8	Med.	T139
149	Claude	Vehicle computer	T	L	Bypass or disabling of DENM signature verification logic, allowing unauthenticated messages to trigger driver warnings.	60.3	High	T164
150	Claude	Vehicle computer	D	L	Flooding with valid-looking DENM messages causes alert fatigue or processing overload, suppressing real warnings.	48.3	Med.	T165
151	Claude	Vehicle computer	E	L	Exploit in warning-logic software grants attacker control over in-vehicle systems beyond the HMI (lateral movement).	60.3	High	T167
152	Claude	Vehicle computer	R	L	No secure logging of received/verified messages, preventing forensic reconstruction of a false-alarm or missed-alarm incident.	13.8	Low	T168
153	Claude	Driver HMI	T	L	Manipulation of alert content/timing to desensitize driver (suppressed alert) or cause distraction (false repeated alerts).	60.3	High	T179
154	Claude	Driver HMI	D	L	Overloading the HMI with spurious alerts to mask a genuine hazard warning.	48.3	Med.	T180
155	Claude	PC5 sidelink	S	A	GNSS/location spoofing combined with message replay to simulate a hazard at a false location.	104.7	Crit.	–
156	Claude	PC5 sidelink	T	A	Message replay attack retransmitting old valid DENM messages to trigger outdated or irrelevant warnings.	104.7	Crit.	link
157	Claude	PC5 sidelink	D	A	RF jamming or protocol-level flooding disrupts V2X communication between RSU and OBU.	52.2	High	link
158	Claude	PC5 sidelink	I	A	Passive sniffing of unencrypted sidelink traffic reveals vehicle movement patterns (privacy/tracking risk).	16.8	Med.	link
159	Claude	Certificates and reports	S	N	Forged or stolen certificate used to authenticate a malicious RSU/OBU to the Central ITS-S.	104.1	Crit.	–
160	Claude	Certificates and reports	T	N	Event reports altered in transit to hide malicious activity or misreport hazard statistics to the operator.	104.1	Crit.	link
161	Claude	Certificates and reports	R	N	Weak non-repudiation controls allow an entity to deny having sent a fraudulent or malformed event report.	32.9	Med.	–
162	Claude	Cameras	S	L	An attacker stages a fake scene in the camera view (e.g., a mannequin on the carriageway, a printed image of an overturned vehicle, or a display placed in front of the lens), so the system reports a non-existent accident and a false DENM is broadcast.	60.3	High	T004
163	Claude	Cameras	T	L	An attacker with physical access to the pole re-aims or moves the camera, so the stored camera position no longer matches the scene and every reported event location is wrong.	60.3	High	T002
164	Claude	Cameras	I	L	Unauthorized access to the camera or its management interface exposes live images of road users, including faces and licence plates (personal data under GDPR).	11.5	Low	T001
165	Claude	Cameras	D	L	The camera is blinded (laser or strong light), covered, painted over or disconnected from power, so real accidents in its field of view are not detected and no warning is issued.	48.3	Med.	T003
166	Claude	Cameras	E	L	Default credentials or vulnerable firmware on the IP infrastructure camera are exploited to gain a foothold on the roadside network, from which the computing unit and the RSU can be attacked.	60.3	High	T005
167	Copilot	Cameras	S	L	Attacker injects fake camera streams or replays previously recorded accident footage to impersonate a live road scene.	60.3	High	T004
168	DeepSeek	Cameras	S	L	Rogue camera node injecting fake video feeds into the edge detector.	60.3	High	T004
169	DeepSeek	Cameras	T	L	Physical or logical modification (e.g., blinding, lens obstruction, replay attacks) of camera streams to prevent accident detection.	60.3	High	T002
170	DeepSeek	Cameras	I	L	Unencrypted video feeds intercepted by unauthorized parties.	11.5	Low	T001
171	DeepSeek	Cameras	D	L	Flooding the camera interface to exhaust processing resources or blinding the sensor.	48.3	Med.	T003